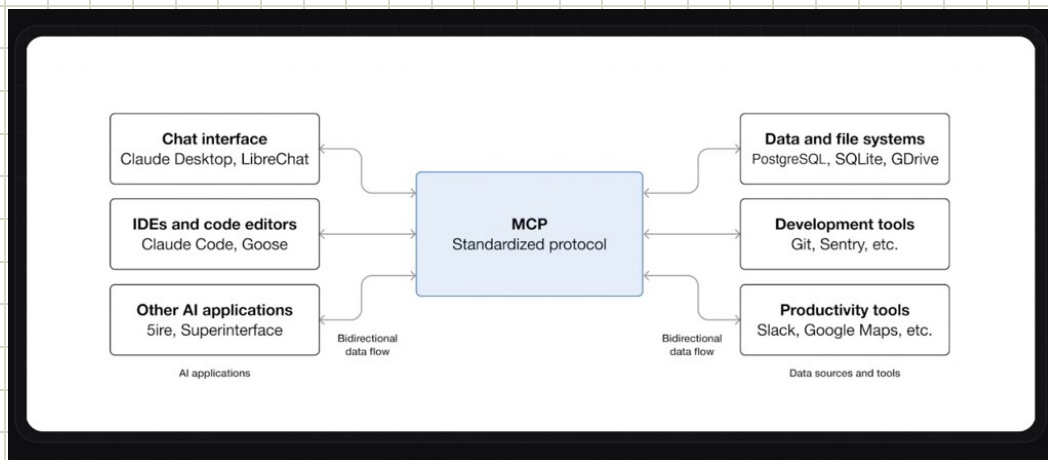


Model Context Protocol!

MCP is open source standard that helps AI applications connect to external systems.

Ex: Claude or ChatGPT can connect to data sources (local files, database), tools (search engines, calculators) and workflows (specialized prompts) enabling them to access key information & perform tasks.



Ex:

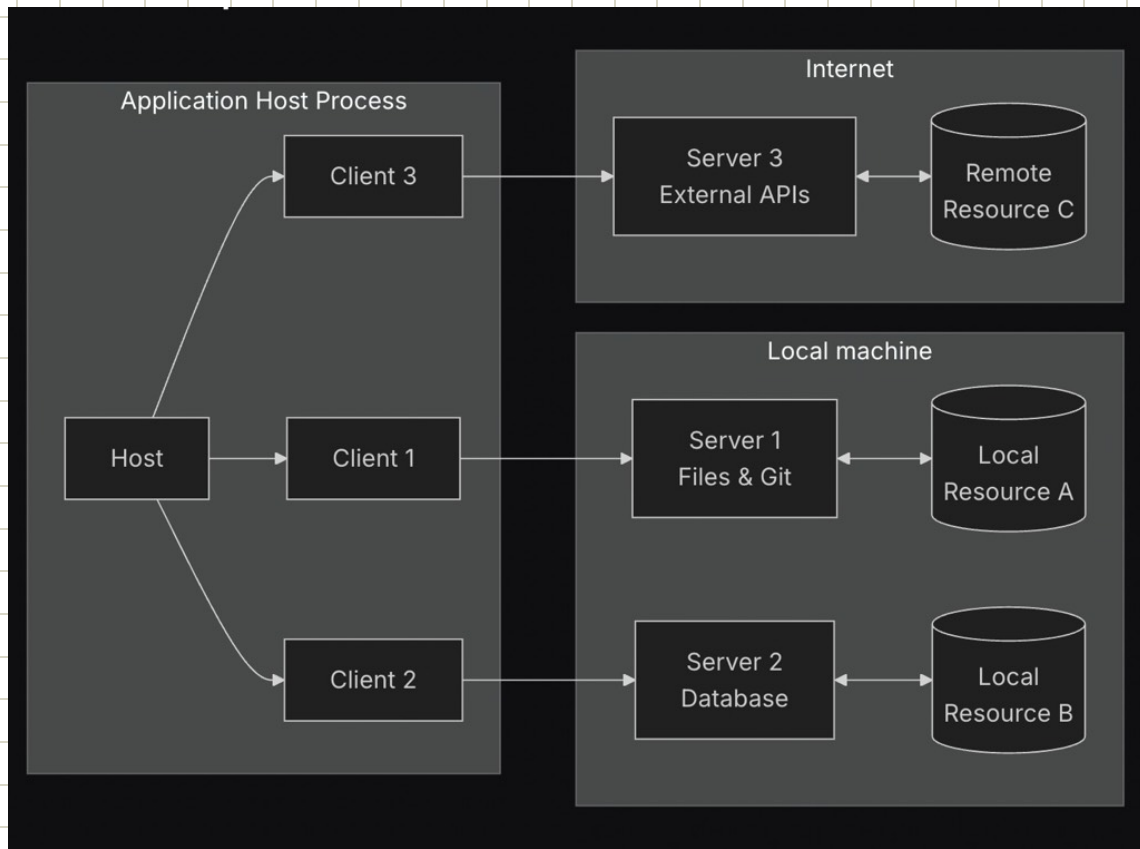
MCP can access Google calendar/Notion and act as a personal AI assistant.

Architecture:

MCP uses client-host-server architecture which enables host to run multiple client instances.

This arch helps to integrate AI capabilities across applications while maintaining clear security boundaries & isolation.

It provides stateful session protocol for context exchanges and sampling coordination between clients & servers.



Host:

- * acts as a container & coordinator
- * creates and manages client instances
- * enforces security policies and consent requirements
- * coordinates AI/ML Integration & sampling
- * Manages context aggregation & isolation.

Client: Each client created by host & maintains an isolated server connection.

- * Creates one stateful session per server.
- * handles protocol negotiations & capability exchange
- * Routes protocol messages bidirectionally
- * Manages subscriptions & notifications
- * Maintains security boundaries between the servers

Host can create multiple clients but each client can have 1:1 relationship with a particular server

Servers:

Servers provide specialized content & capabilities.

- * Expose resources, tools & prompts via MCP primitives.
- * operate independently with focused responsibilities.
- * Request sampling thru client interfaces
- * Can be local/remote services and also must respect security boundaries.

Communication between the host-client-server

Capability Negotiations:

